

NETRA

A First-Principles Network Monitoring Platform

PROJECT SUBMISSION BY:

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Abstract

Netra is an advanced network traffic intelligence platform designed to provide administrators with real-time visibility into local network activity. Unlike traditional tools that rely on heavy third-party dependencies, Netra is built from the ground up using core Python libraries. It combines low-level packet sniffing with a modern web-based dashboard, offering features such as automated service mapping, protocol distribution analytics, and session-based data persistence. A primary focus of the project was the implementation of security measures from first principles, including custom JWT authentication and real-time DoS protection.

1. System Architecture

1.1 Backend Engine

The backend is powered by Flask and utilizes Scapy for raw packet capture. It operates on a multi-threaded architecture where the sniffer runs in a background thread, asynchronously populating a real-time buffer and persisting data to a CSV repository.

1.2 Data Processing

Pandas is utilized for complex data transformations, enabling fast filtering by Protocol, Source IP, or Destination IP. The system also performs automatic 'Well-Known Port' mapping to identify common network services like HTTPS, DNS, and SSH.

2. Security Hardening

Security was a core requirement for this iteration of Netra. The following features were implemented natively:

- Custom JWT Authentication: A first-principles implementation of JSON Web Tokens using HMAC-SHA256 for secure session management.
- User Registration System: Allows multiple users to create accounts with hashed credentials.
- PBKDF2-SHA256 Hashing: Passwords are salted and iteratively hashed to prevent rainbow table attacks.
- Rate Limiting (DoS Protection): A sliding-window algorithm monitors request frequency and

automatically blocks offending IPs.

- Hardened Security Headers: Enforced CSP, X-Content-Type-Options, and HSTS to prevent common web attacks.

3. Core Functionalities

3.1 Live Traffic Dashboard

The dashboard provides a real-time table of incoming and outgoing packets, showing timestamps, IPs, protocols, and packet sizes. It uses a custom AJAX-polling mechanism for seamless updates.

3.2 Statistical Analytics

A dedicated Statistics page visualizes network health through:

- Protocol Distribution (TCP vs UDP vs ICMP)
- Top Source and Destination IP Analysis
- Packet Size Distribution (categorized from Small to Jumbo frames)
- Traffic Intensity Monitoring

4. Deployment Readiness

Netra is optimized for deployment on modern cloud platforms like Render. It includes a Gunicorn Procfile and a comprehensive environment variable configuration for production environments.

5. Conclusion

Netra successfully bridge the gap between network engineering and web application development. By adhering to first principles, the project ensures high performance, low overhead, and maximum security control.